

ANSH MANGUKIYA

AI Engineer · Generative AI · LLM Systems · RAG · Multi-Agent Orchestration

Surat, Gujarat, India | +91 7069216178 | anshmangukiya.ai@gmail.com
anshmangukiya.in | linkedin.com/in/anshmangukiya | github.com/ANSH1370

SUMMARY

AI Engineer with 1.5+ years building production Generative AI systems at Commercient LLC: RAG pipelines, QLoRA fine-tuned LLMs, and multi-agent workflows in LangGraph. Comfortable across the full stack, from multi-GPU vLLM inference to FastAPI services that connect .NET enterprise systems with Python AI. Measured impact so far: **80%** less manual integration coding during customer onboarding, **150+ tokens/sec** inference throughput, and a **60–70%** drop in engineering support and ticket triage effort.

TECHNICAL SKILLS

AI / Machine Learning: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Fine-Tuning (QLoRA / LoRA), Knowledge Distillation, Synthetic Data Generation, RLHF, Prompt Engineering, LLM Evaluation, Inference Optimization, Multi-Agent Systems, AI Agent Orchestration

LLM Frameworks & Tools: vLLM, LangChain, LangGraph, CrewAI, Hugging Face (Transformers, PEFT, TRL, Datasets), Unsloth, BitsAndBytes, PyTorch, Weights & Biases

Vector Databases & Retrieval: Pinecone, Qdrant, ChromaDB, FAISS, Embeddings, Semantic Search

ML & NLP: scikit-learn, TensorFlow / Keras, spaCy

Software Engineering: Python, FastAPI, Flask, Streamlit, .NET Core, C#, ASP.NET, REST API Design, Microservices Architecture, System Design

Cloud & Infrastructure: AWS (EC2, S3), Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, MLOps, Multi-GPU Deployment & Orchestration

Data Engineering & Databases: PostgreSQL, MySQL, DuckDB, SQL Optimization, ETL, Data Pipelines, Data Warehousing, Power BI, Tableau

Specializations: Production LLM Deployment, Enterprise Software Integration, ERP/CRM Systems, Cross-Platform Development (.NET + Python)

PROFESSIONAL EXPERIENCE

AI Engineer — Commercient LLC (Hybrid)

January 2025 – Present

- Architected and deployed a no-code ingestion and chatbot deployment platform enabling enterprise customers to autonomously launch domain-specific GenAI assistants from unstructured data sources (PDFs, web pages, internal documents).
- Built a customer-facing LLM fine-tuning platform: generated **119K+** synthetic training records from customer documents using a self-hosted QwQ-32B teacher model, then trained QLoRA adapters on Llama-3.1-8B and Gemma-2-9B (4-bit NF4, PEFT / TRL, Unsloth, Accelerate) with **300+** runs tracked in Weights & Biases.
- Integrated tuned models into production workflows, reducing manual ticket triage effort by **60%** and improving response accuracy and consistency.
- Engineered a custom multi-GPU LLM inference stack on vLLM with advanced batching, KV-cache optimization, and parameter tuning, achieving **150+ tokens/second** throughput under concurrent load across distributed GPU servers.
- Designed an intelligent SQL view generator using LLM pattern recognition to auto-template ERP-to-CRM data connectors, eliminating **80%** of manual integration coding during new-customer onboarding.

- ▶ Built a conversational ERP-to-CRM sync agent using LangGraph, enabling non-technical users to configure complex integrations via natural language and reducing engineering support load by **70%**.
- ▶ Implemented RAG-based knowledge retrieval pipelines with vector embeddings and semantic search, including a Pinecone-backed relevance gate that keeps off-topic questions away from fine-tuned models after cosine-similarity and TF-IDF gating proved unreliable.
- ▶ Bridged .NET enterprise applications (ASP.NET / C#) with Python AI services through FastAPI microservices and well-defined REST contracts, so data moves reliably between both platforms.
- ▶ Containerized and deployed AI services with Docker on multi-GPU infrastructure, applying CI/CD and MLOps practices for reliable, repeatable production rollouts.

PROJECTS

Production QLoRA Fine-Tuning Pipeline [Write-up ↗](#) | [GitHub ↗](#)

February – July 2025

- ▶ Engineering case study on five months of fine-tuning open models (Llama-3.1-8B, Gemma-2-9B, Qwen2.5-7B) for domain-specific question answering on consumer RTX 4090 GPUs.
- ▶ Covers the full pipeline: synthetic data generation scaled from **2** to **143K** records, a reusable reference trainer (chat templates, early stopping, memory monitoring, multi-GPU Accelerate), and a FastAPI serving layer with a retrieval gate.

Portfolio RAG Assistant [Live on anshmangukiya.in ↗](#) | [GitHub ↗](#)

2026

- ▶ RAG chatbot that answers questions about my work with source citations, embedded as a widget on my portfolio site.
- ▶ LangGraph pipeline (guard → rewrite → retrieve → generate) with bge-small-en-v1.5 embeddings via fastembed (ONNX), Qdrant Cloud for storage, and Llama on Groq using an 8B model for guard / rewrite and a 70B model for generation to keep cost down.
- ▶ Automated retrieval tests gate every deploy at **≥85% hit-rate@4**.

Earlier ML & NLP Projects

2021 – 2025

- ▶ **CareerMatch AI** — Embeds resume skills and job requirements to rank best-fit companies. Python, NLP, scikit-learn.
- ▶ **JobFit ATS Resume Parser** — Parses resumes and scores them against a job description, ATS-style. Python, NLP.
- ▶ **NLP Studio** — Sentiment analysis plus spaCy NER and POS tagging behind a Flask web interface.
- ▶ **SmartPhone Price Intelligence** — Regression price predictor on scraped Smartprix data, with a Streamlit app.
- ▶ **ContextFlow** — LSTM language model that predicts the next word from context. TensorFlow / Keras.

EDUCATION

Bachelor of Engineering — Computer Science

2021 – 2025

Sarvajanik College of Engineering & Technology, Surat, Gujarat

CGPA: 9.78 / 10

CERTIFICATIONS

- ▶ **Get Job Ready: Power BI Data Analytics for All Levels 3.0** — Codebasics [View Certificate ↗](#)
- ▶ **Industrial Artificial Intelligence with Cloud Computing** — Microsoft [View Certificate ↗](#)
- ▶ **Introduction to NLP** — Infosys [View Certificate ↗](#)
- ▶ **Sequences, Time Series and Prediction** — Coursera (DeepLearning.AI) [View Certificate ↗](#)